A TWO-TERM COMPLEX WITH FLAT TERM IN DEGREE ZERO REPRESENTS THE TRUNCATED BASE CHANGE OF ITS DERIVED OBJECT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 22519-22527 (canonical tag 0G9G) and the counted proof is the proof environment at source lines 22529-22550, which immediately follows it and is its own proof.

Editorial note. The proof compares two distinguished triangles and displays the comparison with Xy-pic, using the author's own markup and the shared preamble's array and diagram support.

Editorial note. The chapter file pinned-original-source.tex carries only an input of the shared preamble on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus the native theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key and by this note. The counted statement and proof refer to no labelled item of the chapter and cite no bibliography entry, so this card contributes no external reference stub and needs no bibliography.

Allowed standard prerequisites (editorial).

- A two-term complex concentrated in degrees -1 and 0 sits in a distinguished triangle with the two shifted single-term complexes. used at proof lines 22530-22533, where the triangle K^0 -> K^bullet -> K^{-1}[1] -> K^0[1] is recorded. Provides the diagram of triangles whose base change along R -> R' is compared with the naive tensor product complex.
- For a flat module the derived tensor product with any ring map is the ordinary tensor product, computed in degree zero. used at proof lines 22537-22541 and 22547, where the left and right vertical arrows of the second triangle are isomorphisms because K^0 is flat. Makes the two outer vertical arrows isomorphisms, which by the five lemma on triangles identifies the middle arrow's truncation.
- The map K^{-1} tensor^L R' -> K^{-1} tensor R' is an isomorphism on cohomology in degree 0 and zero on H^{-1}, so the naive tensor complex agrees with the derived one in the degrees relevant to the truncation. used at proof lines 22548-22549. Concludes that the naive tensor product complex represents the truncation of the derived base change.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let R be a ring. Let K be an object of $D(R)$ with $H^i(K) = 0$ for $i \notin \{-1, 0\}$. Let $K^{-1} \to K^0$ be a two term complex of R-modules representing K such that K^0 is a flat R-module (for example projective or free). Let $R \to R'$ be a ring map. Then the complex $K^\bullet \otimes_R R'$ represents $\tau_{\geq -1}(K \otimes^{\mathbf{L}}_R R')$.*

Proof. We have a distinguished triangle

$$K^0 \to K^\bullet \to K^{-1}[1] \to K^0[1]$$

in $D(R)$. This determines a map of distinguished triangles

$$\begin{array}{ccccccc}
K^0 \otimes^{\mathbf{L}}_R R' & \longrightarrow & K^\bullet \otimes^{\mathbf{L}}_R R' & \longrightarrow & K^{-1} \otimes^{\mathbf{L}}_R R'[1] & \longrightarrow & K^0 \otimes^{\mathbf{L}}_R R'[1] \\
\downarrow & & \downarrow & & \downarrow & & \downarrow \\
K^0 \otimes_R R' & \longrightarrow & K^\bullet \otimes_R R' & \longrightarrow & K^{-1} \otimes_R R'[1] & \longrightarrow & K^0 \otimes_R R'[1]
\end{array}$$

The left and right vertical arrows are isomorphisms as K^0 is flat. Since $K^{-1} \otimes^{\mathbf{L}}_R R' \to K^{-1} \otimes_R R'$ is an isomorphism on cohomology in degree 0 we conclude. $\square$